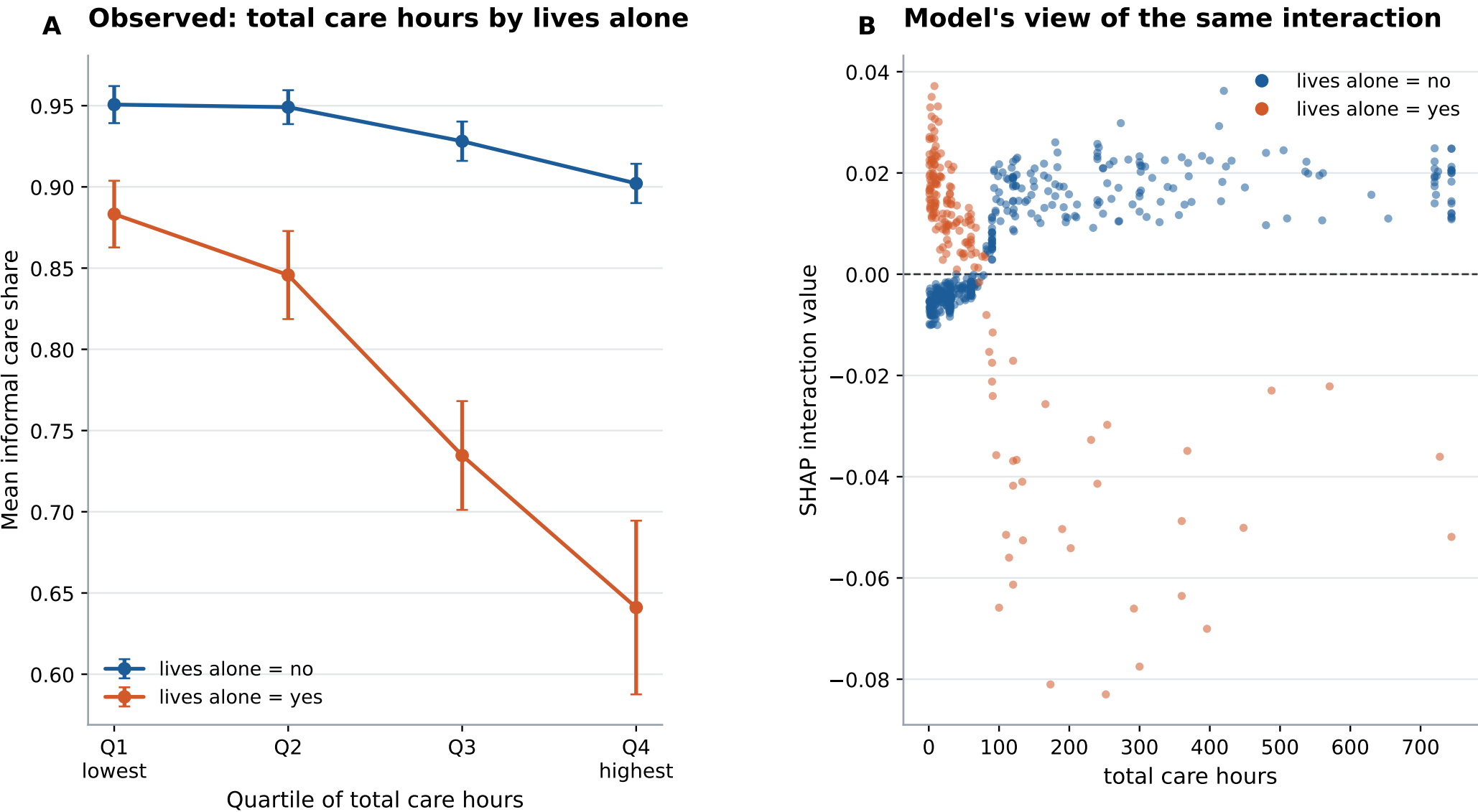


Figure 16. Strongest interaction: total care hours × lives alone ($\beta = -0.032$, $p = 0.005$)



HRS 2022, $n = 2,142$. Panel A splits the sample by lives alone and plots the mean informal share across quartiles of total care hours; non-parallel lines are the interaction. Error bars are standard errors; quartile cells with fewer than 20 respondents are suppressed. Panel B shows the SHAP interaction values for the same pair from the tuned booster, doubled because SHAP splits each pairwise interaction symmetrically between the two features. Agreement between a pre-specified test and a model-derived decomposition is weak evidence that the interaction is real rather than an artifact of either method.